

# CS4084 Research Project Proposal (RP1)

## Performance Analysis of Grover's Search Algorithm Under Realistic Noise Models on NISQ Simulators

### Group Members:

- Saim Haider (22P-9244)
- Muhammad Abdullah (22P-9371)
- Abdullah (22P-9358)

**Instructor:** Dr. Maqsood M. Khan

**Department:** Computer Science

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### Abstract

Grover's search algorithm provides a provable quadratic speedup over classical unstructured search, making it one of the most relevant quantum algorithms for near term deployment. However, current quantum hardware operates in the Noisy Intermediate-Scale Quantum (NISQ) regime, where decoherence and gate errors severely degrade algorithm performance. This research proposes a simulation-based comparative study of Grover's algorithm under three realistic noise models: depolarizing noise, bit-flip errors, and phase-flip errors. Using Python and the Qiskit Aer simulator, Grover's search circuit will be implemented for search spaces of  $n = 2, 3, 4, 5$  qubits. Noise will be injected at parameterized error rates swept across the range  $p \in [0.001, 0.1]$  and success probability will be measured at each configuration. The primary objective is to identify the critical noise threshold  $p^*$  at which Grover's algorithm loses its quantum advantage over classical brute-force search, and to determine how this threshold scales with the number of qubits. Expected outcomes include empirical noise threshold curves for each noise type, a crossover analysis comparing Grover's performance against the classical baseline  $\frac{1}{N}$ , and a reproducible Qiskit codebase suitable for further research. This work directly addresses a gap in the existing literature, where no study has systematically identified this threshold across multiple noise types using modern quantum simulation tools.

## 1 Introduction

Quantum computing leverages the principles of superposition, entanglement, and quantum interference to solve certain computational problems significantly faster than clas-

sical computers. Among the most practically relevant quantum algorithms is Grover’s search algorithm, which achieves  $O(\sqrt{N})$  query complexity for searching an unstructured database of  $N$  entries, compared to the classical  $O(N)$  average case. This quadratic speedup has wide applications in optimization, cryptanalysis, and database search, making Grover’s algorithm a standard benchmark for evaluating near-term quantum hardware.

Despite its theoretical promise, quantum algorithms face a fundamental challenge in practice: noise. Physical qubits are susceptible to environmental decoherence, gate imperfections, and measurement errors. These noise sources collectively corrupt the quantum interference patterns that Grover’s algorithm depends upon to amplify the probability of measuring the correct answer. Present-day quantum processors operate in the Noisy Intermediate-Scale Quantum (NISQ) regime, characterized by gate error rates on the order of  $10^{-3}$  to  $10^{-2}$  and coherence times in the microsecond to millisecond range. Under these conditions, even moderately deep circuits such as Grover’s can suffer severe degradation in success probability.

This research proposes a systematic empirical study of Grover’s algorithm under realistic noise conditions using the Qiskit Aer simulator. By sweeping noise error rates across a controlled range and measuring success probability for each configuration, this work aims to identify the precise noise threshold beyond which Grover’s algorithm loses its quantum advantage over classical search. The findings will provide concrete, reproducible benchmarks directly relevant to the development and calibration of near-term quantum hardware.

## 2 Problem Statement

### What specific problem are you solving?

This research addresses the question: at what noise error rate  $p^*$  does Grover’s search algorithm lose its quantum computational advantage over classical brute-force search, and how does this threshold vary across different noise types (depolarizing, bit-flip, phase-flip) and qubit counts ( $n = 2, 3, 4, 5$ )?

### Why does this problem exist?

Grover’s algorithm relies on precise quantum interference to amplify the target state’s probability amplitude. Any noise that disturbs the phase or amplitude of qubits partially destroys this interference, reducing success probability. As qubit count increases, the circuit depth of Grover’s implementation grows, accumulating more noise over more gate operations. This makes it critical to understand not just that noise degrades performance, but precisely how much noise is tolerable before the algorithm offers no advantage over classical search.

### What are the limitations of existing approaches?

Existing studies have addressed this problem only partially. Salas (2008) derived an

analytical noise threshold using Fortran simulation but studied only depolarizing noise and did not compare against the classical search advantage boundary. Ijaz and Faryad (2023) used Qiskit Aer to study multiple noise types but focused on comparing standard versus QSVT implementations rather than identifying the critical threshold  $p^*$ . Dowarah et al. (2025) derived analytical two-threshold models using Random Matrix Theory but provided no circuit-level simulation. Srivastava et al. (2025) proposed quantum switch mitigation techniques but assumed knowledge of the noise threshold rather than empirically establishing it. No existing study combines a modern Qiskit simulation framework, multiple isolated noise types, a systematic error rate sweep, and explicit identification of the quantum advantage loss threshold.

### 3 Literature Review and Base Papers

#### Ijaz and Faryad (2023) – Base Paper

Ijaz and Faryad conducted a comprehensive noise analysis of Grover’s algorithm using Qiskit Aer for  $n = 3, 4, 5$  qubits with 10,000 shots per configuration. Four noise channels were studied: depolarizing, bit-flip (X), phase-flip (Z), and bit-phase flip (Y). The key finding was that success probability decays exponentially with error probability  $p$  but only linearly with qubit count  $n$ . The bit-phase flip channel caused the most severe degradation across all configurations. This paper is designated as the base paper for our research as it uses the same Qiskit Aer framework and directly characterizes the noise behavior we intend to extend.

**Strength:** Real Qiskit Aer simulation with four noise types; exponential decay relationship confirmed empirically.

**Limitation:** Does not identify the critical threshold  $p^*$  where quantum advantage over classical search is lost; focuses on QSVT comparison rather than threshold analysis.

#### Salas (2008)

Salas performed the earliest systematic numerical study of noise in Grover’s algorithm using Fortran-based Monte Carlo simulation for  $n = 2$  to 7 qubits under a depolarizing channel model. The study derived an exponential damping law for success probability and an allowed-error law:  $\varepsilon_{th}(N) \sim N^{-1.1}$ , where  $N = 2^n$ . An absolute threshold of  $\varepsilon > 0.043$  was identified above which the algorithm fails for any qubit count.

**Strength:** First quantitative noise threshold derived for Grover’s algorithm; foundational analytical results still cited in modern literature.

**Limitation:** Uses an outdated Fortran framework predating Qiskit; studies only depolarizing noise; does not explicitly compare against classical search advantage.

## Dowarah et al. (2025)

Dowarah, Zhang, Khemani, and Kolodrubetz applied Random Matrix Theory (RMT) to model the noisy Grover operator as a Floquet unitary and derived two analytical noise thresholds. The computing transition at  $\delta_{c,\text{comp}} \sim O(2^{-L/2})$  marks where quantum advantage is lost, and the ergodicity transition at  $\delta_{c,\text{gap}} \sim O(L^{-1})$  marks where circuit structure collapses entirely.

**Strength:** State-of-the-art analytical framework; identifies two distinct thresholds with exact scaling laws.

**Limitation:** Entirely theoretical; restricted to systematic coherent noise; no Qiskit simulation or empirical validation provided.

## Srivastava et al. (2025)

Srivastava, Pati, Chakrabarty, and Bhattacharya proposed using quantum switches exploiting indefinite causal order to mitigate depolarizing noise in Grover’s algorithm. Two frameworks were developed, with the second framework (delayed measurement) providing significantly greater noise resilience by preserving correlations between the input state and quantum switch register throughout all iterations.

**Strength:** Novel mitigation approach; demonstrates improved success probability analytically; directly applicable to NISQ devices.

**Limitation:** Entirely analytical with no Qiskit simulation; studies only depolarizing noise; does not identify the threshold  $p^*$  for standard Grover circuits.

## Comparative Analysis

| Paper                       | Method Used                | Strength                                     | Limitation                             |
|-----------------------------|----------------------------|----------------------------------------------|----------------------------------------|
| Ijaz & Faryad (2023) [Base] | Qiskit Aer simulation      | 4 noise types; exponential decay confirmed   | No $p^*$ threshold; QSVT focus         |
| Salas (2008)                | Monte Carlo, Fortran       | First threshold law derived                  | Outdated tools; depolarizing only      |
| Dowarah et al. (2025)       | Random Matrix Theory       | Two-threshold analytical model               | No simulation; coherent noise only     |
| Srivastava et al. (2025)    | Analytical, quantum switch | ICO mitigation; improved success probability | No Qiskit; depolarizing only; no $p^*$ |

## 4 Proposed Methodology

### Proposed Algorithm and System

Grover’s search circuit will be implemented from first principles in Qiskit for  $n = 2, 3, 4, 5$  qubits. Each circuit consists of a Hadamard initialization layer, an oracle operator  $O_\tau = 2|\tau\rangle\langle\tau| - I$  marking the target state, and a diffusion operator  $D = 2|s\rangle\langle s| - I$  performing inversion about the mean. The circuit applies  $k = \lfloor \frac{\pi}{4}\sqrt{N} \rfloor$  Grover iterations to maximize the ideal success probability:

$$P_{\text{success}}(k) = \sin^2\left((2k + 1) \arcsin\left(\frac{1}{\sqrt{N}}\right)\right) \approx 1 \quad (1)$$

### Noise Injection

Three noise models will be injected independently using the Qiskit Aer `NoiseModel` API. Depolarizing noise applies a completely mixed state with probability  $p$ . Bit-flip noise applies a Pauli X gate with probability  $p$ . Phase-flip noise applies a Pauli Z gate with probability  $p$ . Error rates will be swept across  $p \in [0.001, 0.1]$  in steps of 0.005 and each configuration will be executed with 1024 shots averaged over 5 independent runs.

### Threshold Identification

The critical threshold  $p^*$  is defined as the error rate at which Grover’s measured success probability crosses the classical random search baseline  $\frac{1}{N} = \frac{1}{2^n}$ . This crossover

point will be identified numerically for each combination of noise type and qubit count.

### **Innovation Over Base Papers**

This study extends Ijaz and Faryad (2023) by restricting analysis to the standard Grover implementation and explicitly locating  $p^*$ , which their work does not provide. It extends Salas (2008) by using modern Qiskit tooling and testing multiple noise channels. It provides the empirical foundation that Srivastava et al. (2025) presuppose but do not establish.

## **5 Tools and Implementation Plan**

- **Programming Language:** Python 3.10+
- **Quantum Framework:** Qiskit 1.x with Qiskit Aer noise simulator
- **Noise API:** `qiskit_aer.noise.NoiseModel`
- **Visualization:** matplotlib, numpy
- **Platform:** IBM Quantum Lab (free tier) or local Aer simulator
- **No external dataset required:** The search space is synthetically generated as an  $n$ -qubit register

## **6 Expected Results**

### **What results do you expect?**

We expect success probability to decay monotonically as error rate  $p$  increases for all noise types. Based on the literature, phase-flip noise is expected to cause more severe degradation than bit-flip noise for Grover circuits due to the algorithm’s sensitivity to phase interference. Depolarizing noise is expected to cause the earliest threshold crossing as it simultaneously corrupts both phase and amplitude information.

### **How will performance be measured?**

Performance will be measured as the probability of measuring the correct target state across 1024 shots per configuration. The primary output will be a family of success probability versus error rate curves, one per noise type per qubit count, with the threshold  $p^*$  marked on each curve. Secondary metrics will include circuit depth and gate count per qubit configuration.

### **What improvement do you anticipate compared to base papers?**

Compared to Ijaz and Faryad (2023), this work will produce an explicit quantum advantage threshold map that their study does not provide. Compared to Salas (2008), results will be validated on a modern Qiskit circuit simulator with three noise types rather than one. The threshold values obtained will also serve as a direct empirical input for

noise mitigation strategies such as the quantum switch framework of Srivastava et al. (2025).

## 7 Proposed Timeline

- **Week 1 to 2:** Literature review and circuit design for all qubit configurations
- **Week 3 to 5:** Qiskit circuit implementation and noiseless baseline validation
- **Week 6 to 7:** Noise model integration, error rate sweep, and data collection
- **Week 8:** Threshold analysis, plot generation, and IEEE paper writing

## References

- [1] L. K. Grover, “A fast quantum mechanical algorithm for database search,” in *Proc. 28th Annual ACM Symposium on Theory of Computing (STOC)*, 1996, pp. 212–219.
- [2] M. A. Ijaz and M. Faryad, “Noise analysis of Grover and phase estimation algorithms implemented as quantum singular value transformations for a small number of noisy qubits,” *Scientific Reports*, vol. 13, p. 20144, 2023.
- [3] P. J. Salas, “Noise effect on Grover algorithm,” *The European Physical Journal D*, vol. 46, pp. 365–381, 2008.
- [4] S. Dowarah, C. Zhang, V. Khemani, and M. Kolodrubetz, “Phases and phase transition in Grover’s algorithm with systematic noise,” *arXiv preprint arXiv:2406.10344v2*, 2025.
- [5] S. Srivastava, A. K. Pati, I. Chakrabarty, and S. Bhattacharya, “Using quantum switches to mitigate noise in Grover’s search algorithm,” *arXiv preprint arXiv:2401.05866v2*, 2025.